

Heechan Yi

☎ +82) 10-7126-0640 ✉ lhc320@yonsei.ac.kr 🌐 <https://github.com/HeechanYi>

EDUCATION

Yonsei University, Seoul <i>Integrated M.S.–Ph.D. Program in Physics</i>	Sep. 2025 ~ Present <i>Yonsei HEP–Cosmo</i>
Konkuk University, Seoul <i>B.S. dual major in Physics & Mathematics</i>	Mar. 2019 ~ Aug. 2025 <i>Total GPA: 4.26/4.5 (Physics: 4.46/4.5, Mathematics: 4.35/4.5)</i>

RESEARCH INTERESTS

- **High Energy Physics & Cosmology** (Particle Phenomenology, Early Universe, Gravitational Waves)
- **Quantum Computation & Sensing** (Quantum Simulation, Quantum Sensing for High Energy Physics)
- **AI for Physics**

PUBLICATIONS

- [2608.02733] **High-Frequency Gravitational Wave Detection with Superconducting Qubits**
 - **Heechan Yi**, Seong Chan Park, Kyoungchul Kong, Myeonghun Park
- [2510.10501] **Quantum Integration Networks for Efficient Monte Carlo in High-Energy Physics**
 - **Heechan Yi**, Kayoung Ban, Myeonghun Park, Kyoungchul Kong
- [2410.22417] **Hybrid quantum-classical approach for combinatorial problems at hadron colliders**
 - Jacob L. Scott, Zhongtian Dong, Taejoon Kim, Kyoungchul Kong, Myeonghun Park, **Heechan Yi**, Willie Aboumrads, Ananth Kaushik, Martin Roetteler

RESEARCH PROJECTS

- High-Frequency Gravitational Wave Detection with Superconducting Qubits** | *Quantum Sensing, HEP* 2025 ~ Present
 - Developed a quantum-sensing framework for detecting GW-induced electromagnetic signals using superconducting qubits in resonant microwave cavities
 - Investigated localized qubit readout and collective quantum enhancement for high-frequency gravitational-wave searches
- Quantum Integration Networks for Efficient Monte Carlo in HEP** | *QML, HEP* 2024 ~ 2025
 - Developed quantum-circuit-based numerical integration methods for Monte Carlo applications in high-energy physics
 - Investigated data re-uploading, sampling strategies, and quantum circuit architectures for efficient numerical integration
- Quantum Algorithms for Collider Combinatorics** | *Quantum Computation, Collider Physics* 2024 ~ 2025
 - Contributed the variational quantum imaginary-time evolution (VarQITE) analysis to a collaborative study of quantum algorithms for combinatorial reconstruction at hadron colliders

TALKS & PRESENTATIONS

Invited Talks

PARTIQAL 2026 | Jul. 2026

- High-Frequency Gravitational Wave Detection with Superconducting Qubits

CTPU-CGA-Yonsei Joint School and Workshop | Feb. 2026

- High-Frequency Gravitational Wave Detection with Superconducting Qubits

The 2nd AI+HEP in East Asia Workshop | Jan. 2026

- Quantum Integration Networks for Efficient Monte Carlo in High-Energy Physics

The 22nd Yonsei-Saga Partnership Program | Oct. 2025

- Quantum Approach to High Energy Physics – Quantum Sensing

Poster Presentations

IAIFI Summer Workshop 2026 | Aug. 2026

- Quantum Integration Networks for Efficient Monte Carlo in High-Energy Physics

2025 KPS Spring Meeting | Apr. 2025

- Quantum Integration Networks for Efficient Monte Carlo in High-Energy Physics

AWARDS & HONORS

Awards

- Dean's List for 5 semesters (Fall 2024 with GPA 4.35/4.5, Spring 2024 with GPA 4.5/4.5, Fall 2023 with GPA 4.25/4.5, Spring 2023 with GPA 4.41/4.5, Fall 2022 with GPA 4.42/4.5)

Scholarships

- Master's Excellence Scholarship in Science and Engineering, Korea Student Aid Foundation (KOSAF), 2026–Present
- Received Academic Achievement Scholarship from Woon Hea Foundation in 2024
- Earned merit-based tuition discount(40% ~ 100%) for 7 semesters from Konkuk University

ACADEMIC EXPERIENCES

Advanced School Program

CANU Summer School 2024 - Read up on cosmic particles, neutrino physics and its experiments

CPTU-IBS Winter School 2024 - Studied basic cosmology and gravitational waves

LSSU Winter School 2024 - Learned the Standard Model and the Higgs mechanism

QI Winter School 2024 - Studied machine learning and quantum machine learning

KIAS Particle Physics Summer Camp 2023 - Studied basic collider physics and dark matter

APCTP LHC-Physics Workshop | *Student Assistant*

Aug. 2019 & Aug. 2022

TECHNICAL SKILLS

Programming

: Python (NumPy, SciPy, Matplotlib etc.), Git/GitHub, Jupyter

High Energy Tools

: MadGraph, Pythia8, SOFIA

Quantum Computing & Machine Learning

: PennyLane, Qiskit, PyTorch, Scikit-learn

Speaking Languages

: Korean (native), English (intermediate)

EXTRACURRICULAR ACTIVITIES

Military Service

Feb. 2020 ~ Aug. 2021

Honorably discharged as a sergeant in Defense Counterintelligence Command, Korea